UE26CY151A Engineering Chemistry · Unit 2 · Electrochemistry and Corrosion

# Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

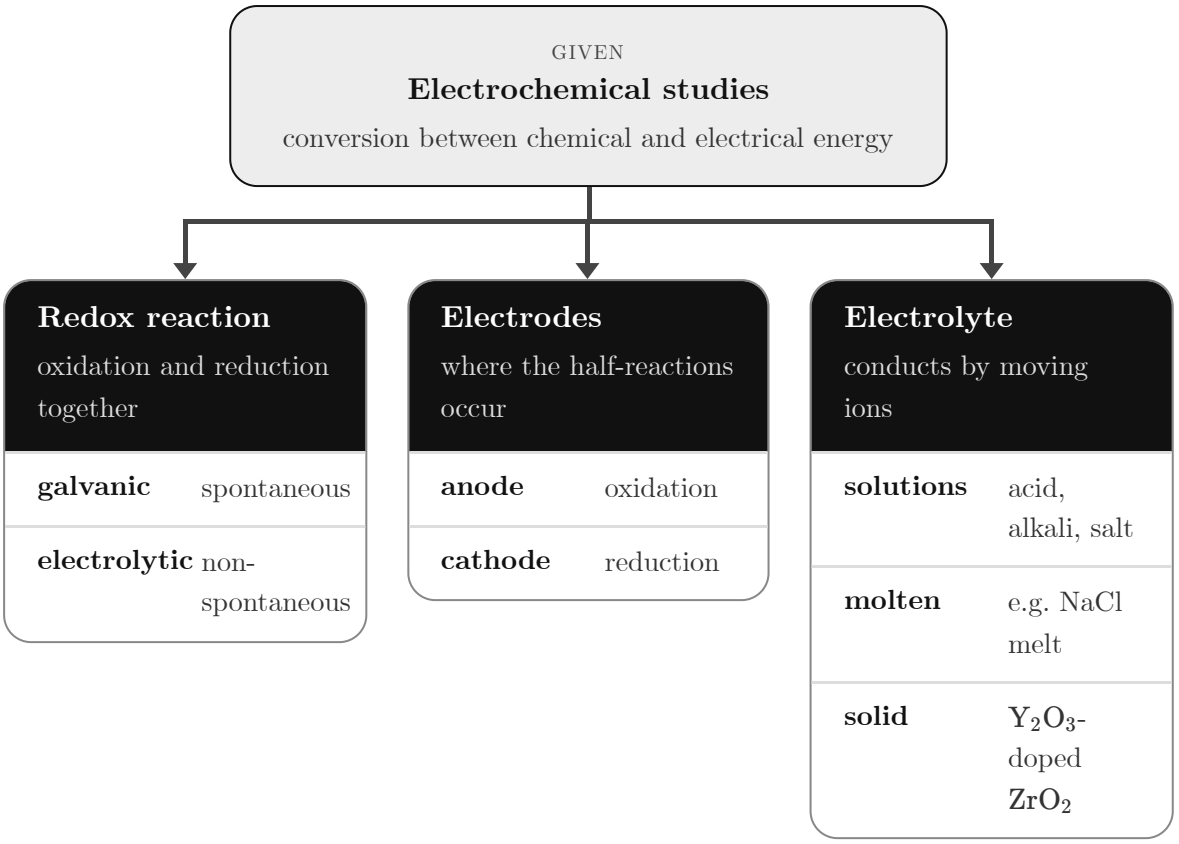
Class 001 · Origin of electrode potential, cell potential · compiled 17 October 2026

| | |
|---|---|
| ESA ★★★ | Asked repeatedly in past ESA papers |
| ESA ★★ | Asked in past ESA papers or computed in class-work problems |
| ESA ★ | Conceptual: taught in class, not examined so far |
| ADDED FOR CLARITY | Standard context that is not in the class material |
| TRANSCRIPT ONLY | Said in the lecture but not on any slide |
| VIDEO ONLY | Seen only in a frame of the class video |
| NOT ON CURRENT SLIDES | From older material or past papers, not this year's slides |
| S001·10 | Deck 001, slide 10 (rendered .pptx) |
| T001 00:08:22 | AV-summary transcript 001, segment start |
| ESA Jan-25 s2 2b | Past paper: Jan 2025 set 2, question 2.b |
| ESA Jul-23a 4d | Past paper: Jul 2023 (first UE22 paper), question 4.d |

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# 1 Electrochemistry and the two types of cell




***Remark.*** The molten and solid examples come from the lecture (T001 00:04:28–00:05:32).

**Figure F1.1.** What every electrochemical system needs ESA ★

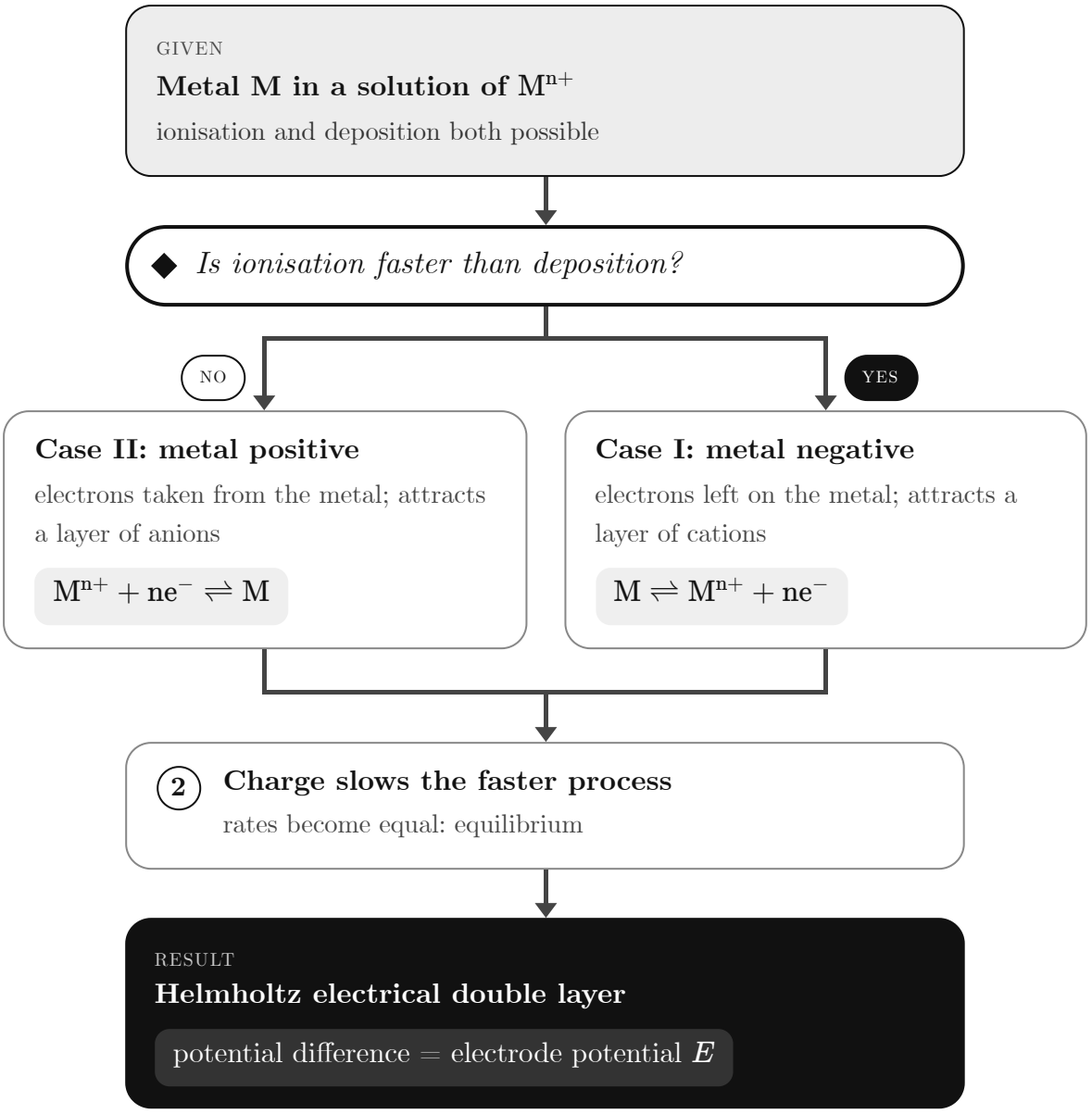
S001·4 · T001 00:02:58

## Galvanic cell versus electrolytic cell

| | Galvanic cell | Electrolytic cell |
|---|---|---|
| **Energy conversion** | chemical → electrical | electrical → chemical |
| **Redox reaction** | spontaneous | non-spontaneous (driven) |
| **$\Delta G$, $E_{cell}$** | $\Delta G < 0$, $E_{cell} > 0$ | $\Delta G > 0$, $E_{cell} < 0$ for the driven reaction |
| **Anode** | oxidation; negative terminal | oxidation; connected to + of the supply |
| **Cathode** | reduction; positive terminal | reduction; connected to − of the supply |
| **Salt bridge** | usually two solutions joined by a salt bridge | usually one vessel, no salt bridge |
| **Examples** | batteries, fuel cells, Daniell cell | electroplating, electrorefining, metallurgy |

added for clarity S001·3 · T001 00:01:15 · T001 00:02:19

# 2 Electrode potential and its origin: the Helmholtz double layer




***Exam trap.*** Case I: metal (−), solution layer (+). Do not swap them.

**Figure F2.1.** How the Helmholtz double layer forms ESA ★★★

S001·5 · S001·6 · S001·7 · ESA Jan-25 s2 2b

## Case I versus case II of the double layer

| | Case I | Case II |
|---|---|---|
| **Faster reaction** | ionisation (oxidation) $M \rightarrow M^{n+} + ne^-$ | deposition (reduction) $M^{n+} + ne^- \rightarrow M$ |
| **Charge on metal** | net negative | net positive |
| **Layer attracted** | positive ions | negative ions |
| **Effect of the charge** | retards ionisation, increases deposition | retards deposition, increases ionisation |
| **Result** | equilibrium + Helmholtz double layer | equilibrium + Helmholtz double layer |
| **Typical metals (added for clarity)** | active metals, e.g. Zn | less active metals, e.g. Cu |

S001·6 · S001·7

# 3 Standard electrode potential, electrochemical cell and cell notation

### Four potentials compared

| Quantity | Symbol | What it is | Conditions |
|---|---|---|---|
| **Electrode potential** | $E$ | potential at one metal–solution interface | any; not measurable alone |
| **Standard electrode potential** | $E^\circ$ | same, under standard conditions | 1 M, 298 K, 1 atm (gases) |
| **Cell potential (emf)** | $E_{cell}$ | difference of the two electrode potentials | any concentrations |
| **Standard cell potential** | $E^\circ_{cell}$ | $E^\circ_{cathode} - E^\circ_{anode}$ | 1 M, 1 atm, 298 K |

S001·4 · S001·8 · S001·9

# 4 Cell potential, standard cell potential and free energy

GIVEN

**Two half-cells and their $E^\circ$ values**

reduction potentials (series or question)

↓

**1** **Lower $E^\circ$ = anode, higher = cathode**

anode is oxidised; write it on the left

↓

**2** **Write the notation and half-reactions**

anode | ion || ion | cathode

↓

**3** **Balance electrons**

$n$ = electrons in the overall reaction

e.g. Fe/Au: $n = 2 \times 3 = 6$

↓

**4** **Subtract (no coefficients on $E^\circ$)**

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$$

↓

**5** **Free energy**

$$\Delta G^\circ = -nFE^\circ_{cell},\ F = 96500$$

↓

RESULT

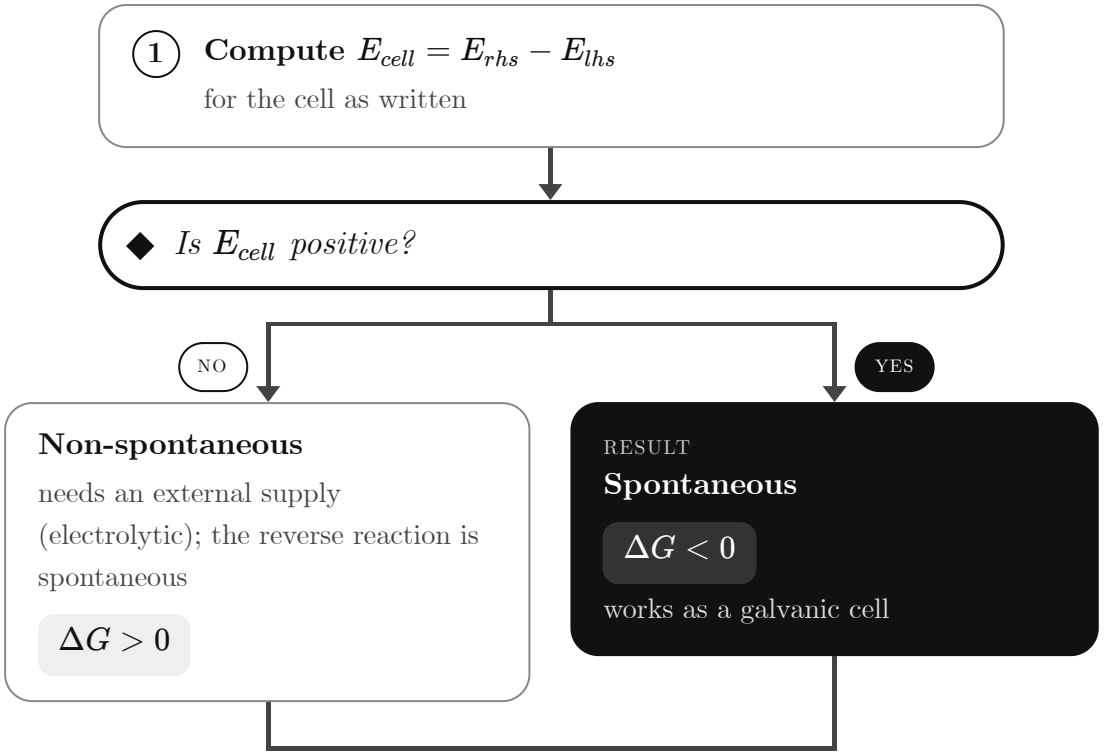
**Interpret**

$E^\circ_{cell} > 0,\ \Delta G^\circ < 0$: spontaneous

***Remark.*** For non-standard concentrations continue with the Nernst equation (class 002); the ESA cell numericals (session 006) all start with steps 1–4.

**Figure F4.1.** From two electrodes to $E^{\circ}_{cell}$ and $\Delta G^{\circ}$ ESA ★★★

S001·8 · S001·9 · S001·10




***Remark.*** $\Delta G = -nFE_{cell}$: $n$ and $F$ are positive, so $\Delta G$ and $E_{cell}$ always have opposite signs.

**Figure F4.2.** Spontaneous or not? ESA ★★★

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# 5 The electrochemical series and its uses

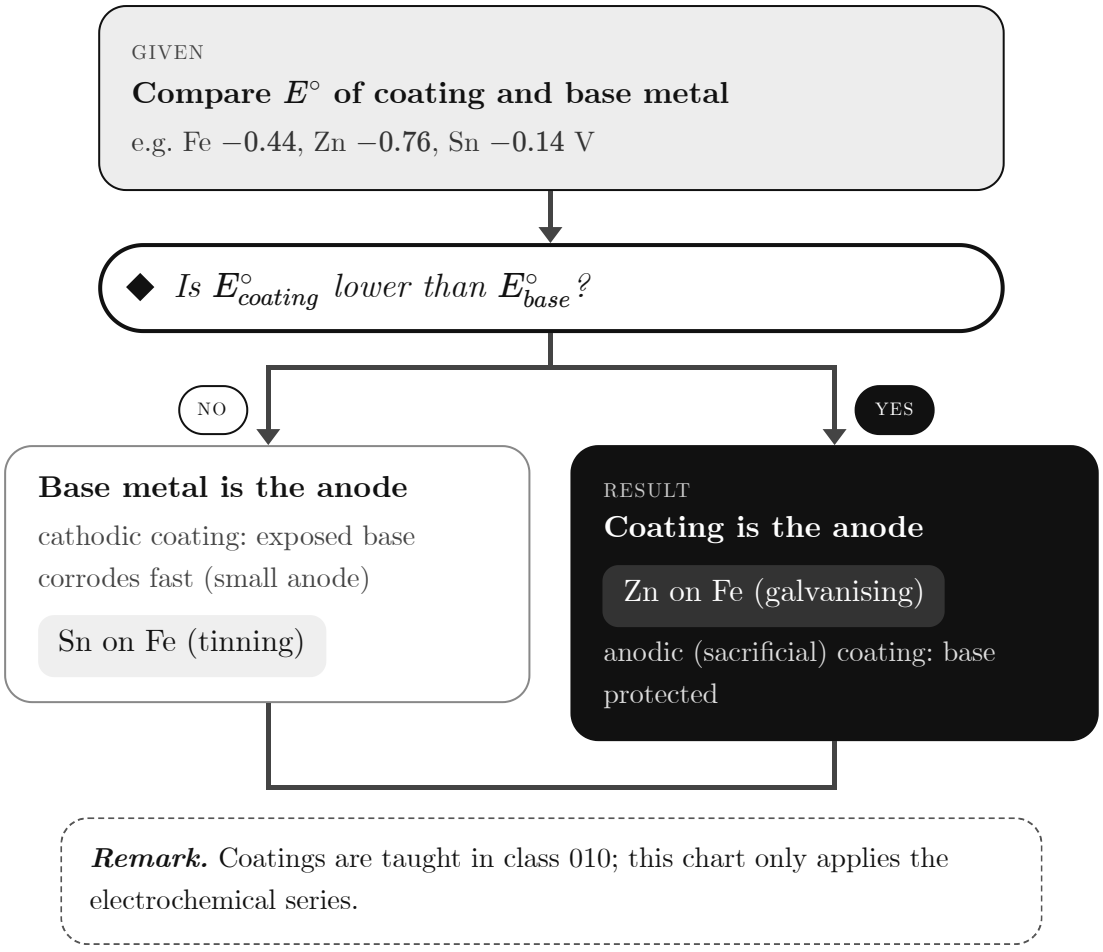



**Figure F5.1.** Does the base metal corrode when the coating is scratched? ESA ★★

ESA Jul-23a 4d · S001·10

### Electrochemical series (slide 001·10)

| Couple (oxidant ⇌ reductant) | $E^\circ$ (V) | Tendency |
|---|---|---|
| $Li^+ + e^- \rightleftharpoons Li$ | −3.03 | strongest reducing metal |
| $K^+ + e^- \rightleftharpoons K$ | −2.92 | oxidation |
| $Ca^{2+} + 2e^- \rightleftharpoons Ca$ | −2.87 | oxidation |
| $Na^+ + e^- \rightleftharpoons Na$ | −2.71 | oxidation |
| $Mg^{2+} + 2e^- \rightleftharpoons Mg$ | −2.37 | oxidation |
| $Al^{3+} + 3e^- \rightleftharpoons Al$ | −1.66 | oxidation |
| $Zn^{2+} + 2e^- \rightleftharpoons Zn$ | −0.76 | oxidation |
| $Fe^{2+} + 2e^- \rightleftharpoons Fe$ | −0.44 | oxidation |

| Couple (oxidant ⇌ reductant) | $E^\circ$ (V) | Tendency |
|---|---|---|
| $Pb^{2+} + 2e^- \rightleftharpoons Pb$ | $-0.13$ | oxidation |
| $2H^+ + 2e^- \rightleftharpoons H_2$ | 0.00 | reference (SHE) |
| $Cu^{2+} + 2e^- \rightleftharpoons Cu$ | $+0.34$ | reduction |
| $Ag^+ + e^- \rightleftharpoons Ag$ | $+0.80$ | reduction |
| $Au^{3+} + 3e^- \rightleftharpoons Au$ | $+1.50$ (ESA: 1.52) | strongest oxidising ion |

S001·10

### Zinc coating versus tin coating on iron

| | Galvanising (Zn) | Tinning (Sn) |
|---|---|---|
| **$E^\circ$ of coating vs Fe ($-0.44$ V)** | $-0.76$ V: lower | $-0.14$ V: higher |
| **At a scratch, the anode is** | zinc | iron |
| **Type of coating** | anodic (sacrificial) | cathodic |
| **Iron at the scratch** | protected | corrodes rapidly (pitting) |

ADDED FOR CLARITY ESA Jul-23a 4d · S001·10